Robustness of carryover-mitigation analysis strategies for biomarker-treatment interaction: a factorial simulation study

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Contents

1 Abstract

Background. Carryover effects in aggregated N-of-1 and crossover trials degrade the detectability of biomarker-treatment interactions. Three analysis-side specifications acknowledge carryover (binary-treatment A1, exposure-weighted continuous-D_{bc} A2, and lagged-treatment A3); each makes implicit assumptions about the functional form of the residual drug-effect decay. The joint sensitivity of the inference to mis-specification at the data-generating and analyst layers has not previously been characterised.

Methods. We conduct an ADEMP-compliant factorial simulation study[1] crossing three data-generating carryover decay forms (linear, exponential, Weibull) against the three analysis-model specifications, under both mean-moderation (Architecture~A) and multivariate-normal differential-correlation (Architecture~B) biomarker data-generating processes. The principal factorial covers 540 cells at 500 replicates per cell. Five sensitivity blocks (S1-S5) vary autocorrelation strength, analyst-versus-DGP half-life mismatch, dropout rate, between-subject heterogeneity, and biomarker correlation strength. All performance measures are reported with Monte Carlo standard errors per Morris et al. (2019).

Results. Across the principal factorial and all sensitivity blocks, the exposure-weighted A2 specification consistently produced the highest power, with an absolute advantage of approximately 10 percentage points over A1 at the reference cell (Architecture~B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ week; A2 $= 0.860$, A1 $= 0.766$, A3 $= 0.770$; approximately five MCSE of separation). The ranking A2 $>$ A1 $\approx$ A3 was preserved across designs, architectures, sample sizes, and DGP carryover half-lives. A2 retained its advantage when the analyst-assumed half-life departed from the true half-life by a factor of two or four. A high-precision rerun at $N \in \{35, 70\}$ and $t_{1/2}^{\mathrm{DGP}} \in \{0.5, 1.0\}$ confirmed the ranking at MCSE 0.020 with a paired McNemar test $p = 6.6 \times 10^{-17}$ at the highest-leverage cell.

Conclusions. A2 is the recommended default for the analysis of biomarker-

treatment interactions in aggregated N-of-1 trials under realistic carryover. A3 is a defensible fallback when the pharmacokinetic half-life is unknown but loses approximately 30 percent of A2's advantage at larger N. A1 (the strict-RM-ANOVA analogue) is dominated across the entire grid.

2 Introduction

We begin with a question that has received surprisingly little systematic attention in the biomarker-moderated N-of-1 trial literature. When residual drug effect persists after the on-drug phase, any analysis that ignores carryover or mis-specifies its functional form will, in principle, lose power to detect the biomarker-treatment interaction. But how large, in practice, is that loss, and can it be recovered by a better-chosen analysis specification? To the best of our knowledge, no systematic evaluation of this question exists for the within-subject biomarker-interaction setting.

The companion manuscript at `analysis/report/01-dgp-mean-moderation-vs-mvn/` establishes one part of the picture. Hendrickson[2] introduced the biomarker-by-treatment interaction framework for aggregated N-of-1 trials and reported power under a matched exponential decay model. The companion paper extended that work by comparing two data-generating architectures (direct mean moderation under Architecture A, and multivariate-normal differential correlation under Architecture B) and found that the power consequences of carryover differ qualitatively between them. Under Architecture B, carryover at a half-life of one week reduces power by 40 to 60 percentage points from the no-carryover baseline; under Architecture A the corresponding loss is 8 to 13 percentage points. That prior work, however, held the decay form fixed at exponential and evaluated a single analysis-model specification throughout. Unfortunately, the joint sensitivity of the inference to mis-specification at both the data-generating and analyst layers remained uncharacterised.

The present paper addresses that gap directly. The trial designer must specify a carryover decay form, explicitly or implicitly, and the analyst must choose how to represent residual drug exposure in the linear predictor. In practice, the true pharmacokinetic decay is rarely known with precision at the design stage: the analyst typically defaults to a convenient parametric form or ignores carryover entirely. We have not, to the best of our knowledge, seen a systematic evaluation of which analysis-side specification holds up best when the assumed decay form departs from truth, and it is that question we shall take up here.

To address this, we perform a factorial simulation study crossing three DGP decay forms (linear, exponential, and Weibull) against three analysis-model carryover specifications: a binary on-drug indicator that ignores carryover altogether (A1), an exposure-weighted continuous predictor that commits to a specific decay form and half-life (A2, the current pmsimstats-ng default), and a lagged-treatment indicator that estimates carryover magnitude without positing a functional form (A3, following the classical crossover-trial approach surveyed by Jones and Kenward[3]).

The full grid is run under both DGP architectures to establish whether robustness rankings are preserved across data-generating mechanisms or whether the preferred specification is architecture-dependent.

The study is organized and reported following the ADEMP framework of Morris, White, and Crowther[1], which structures simulation studies around aims, data-generating mechanisms, estimands, methods, and performance measures. Each performance measure is reported per cell alongside its Monte Carlo standard error, as Morris et al. recommend. We proceed in §3 through the simulation design in ADEMP order; results are presented in §4; and §5 takes up the practical implications for trialists choosing among the three carryover specifications.

3 Simulation design

This section organises the study under the ADEMP labels of Morris et al.[1]. We begin in §3.1 with the data-generating mechanisms, then turn in §3.2 to the estimands, §3.3 to the methods (analysis-model specifications), §3.4 to the factorial grid, and §3.5 to the parameter settings and performance measures. Reproducibility metadata, including the R version, key package versions, the RNG seed strategy, and elapsed time, are recorded in §3.6 (Computing environment).

3.1 Aims

The primary aim of this study is to quantify the cost of analysis-model carryover mis-specification in aggregated N-of-1 and crossover trials, and to identify which mitigation strategy is most robust across the DGP decay-form grid. A secondary aim is to verify that the robustness rankings transfer across the two data-generating-process architectures (mean moderation versus differential correlation), so that a practitioner can be reasonably confident that the recommendation does not depend on which biomarker biology is assumed.

3.2 Data-generating mechanisms

We retain both DGP architectures from the companion manuscript. In Architecture A (direct mean moderation), the biomarker B_i enters the mean structure as $Y_{it} = \mu_0 + \beta_1 D^*_{it}(1 + \beta_{bm} B_i) + \epsilon_{it}$, where D^*_{it} is the effective exposure at timepoint t under the chosen decay form. In Architecture B (MVN differential correlation), the biomarker and biological response are jointly drawn from a multivariate normal with treatment-state-dependent correlation $\mathrm{Cor}(B_i, BR_{it}) = c_{bm} \cdot \phi(t_{sd})$, where ϕ is the decay form.

In both architectures the decay function $\phi : \mathbb{R}_+ \to [0, 1]$ maps time since drug discontinuation t_{sd} to a residual-effect multiplier. We evaluate three forms, each parameterised to match a common half-life $t_{1/2}$:

3.2.1 Linear decay

$$\phi_{\text{lin}}(t_{sd}) = \max\left(0,\, 1 - \frac{t_{sd}}{2t_{1/2}}\right)$$

Residual effect vanishes completely at $t_{sd} = 2t_{1/2}$. The linear form is biologically implausible for drugs with first-order elimination kinetics, but it represents a defensible pragmatic assumption when the only available information is that carryover has fully worn off by some specified time X.

3.2.2 Exponential decay

$$\phi_{\text{exp}}(t_{sd}) = \exp\left(-\lambda t_{sd}\right), \qquad \lambda = \frac{\ln 2}{t_{1/2}}$$

This is the form used throughout the companion manuscript and currently implemented in `implementations/tidyverse/R/functions.R`. It corresponds to first-order elimination kinetics and is the default assumption in classical pharmacokinetics; we retain it here as the natural reference against which the linear and Weibull forms are compared.

3.2.3 Weibull decay

$$\phi_{\text{weib}}(t_{sd}) = \exp\left(-\left(\lambda_w t_{sd}\right)^k\right), \qquad \lambda_w = \frac{(\ln 2)^{1/k}}{t_{1/2}}$$

A shape parameter k governs whether the elimination tail is heavier ($k < 1$, slow elimination) or lighter ($k > 1$, accelerated washout) than the exponential, which is recovered exactly at $k = 1$. We report results at $k = 0.7$, representing a drug with a prolonged low-concentration tail, and at $k = 1.5$, which captures drugs with saturable clearance or active-transporter kinetics that clear more rapidly once the concentration falls below the saturation threshold.

Implementation note. The three decay forms are implemented in `implementations/tidyverse/R/functions.R` via the `carryover_decay()` helper function, with a `carryover_form` and `weibull_shape` pair of `model_param` fields plumbed through `apply_carryover_to_component()`, `build_correlation_matrix()`, and `build_sigma_matrix()`. Backward compatibility is preserved by defaulting to the exponential form when `carryover_form` is absent. We note that the `implementations/original-extended/` collection has not been updated and supports only the exponential form; cross-validation across implementations is therefore limited to exponential cells, which is sufficient for pipeline verification but not for the full decay-form comparison.

3.3 Estimands

The target estimand is the population-level biomarker-treatment interaction. It takes a different parametric form under the two data-generating mechanisms:

- **Architecture A.** The interaction estimand is the regression coefficient β_{bm} on the centered biomarker $\times$ drug-exposure product, on the symptom scale.
- **Architecture B.** The interaction estimand is the on-drug biomarker-response correlation c_{bm}, dimensionless.

The two estimands are not directly commensurable. To enable cross-architecture comparison of bias and power, we calibrate both architectures to a common population effect-size scale, defined as the expected difference in mean on-drug response per unit biomarker standard deviation at $t_{1/2} = 0$. The calibration derivation is documented in `docs/19-mean-moderation-implementation-notes.tex`. Bias and empirical standard error in the Results section are reported on this calibrated scale; power and Type I error are reported on the native test-statistic scale.

The null estimand for Type I error verification is $\beta_{bm} = 0$ under Architecture A and $c_{bm} = 0$ under Architecture B, each realised by the $c_{bm} = 0$ row of the parameter grid.

3.4 Methods: analysis-model carryover specifications

The analysis model is `nlme::lme(Sx ~ bm * X + t, random = ~1 | ptID, correlation = corCAR1(form = ~ t | ptID))` for all specifications. The three specifications differ in the drug exposure predictor X:

3.4.1 A1: binary-treatment (carryover ignored)

$X = D_{it}$, the on-drug indicator. This is the naive specification that disregards any residual drug effect during washout; it serves as the reference against which the power cost of ignoring carryover is measured.

3.4.2 A2: exposure-weighted (matched-decay continuous predictor)

$X = \mathrm{Dbc}_{it}$, the continuous exposure-decayed predictor constructed to match the analyst's assumed decay form and half-life. This is the current pmsimstats-ng default specification. When the analyst's assumed form coincides with the true DGP form the predictor is well-matched; when it does not, the predictor is misspecified in a way that the Tier 2 block S2 is designed to probe.

3.4.3 A3: lagged-treatment (estimated carryover term)

$X = D_{it}$ augmented with a lagged-treatment indicator $L_{it} = D_{i,t-1}(1 - D_{it})$ that marks off-drug timepoints following an on-drug timepoint. The carryover magnitude is estimated from the data rather than assumed a priori, so the specification does not require the analyst to posit a functional decay form. This approach follows the classical crossover-trial framework surveyed in[3].

Table 1: Illustrative D_{bc} values at three off-drug timepoints under three analyst-assumed half-lives, exponential decay, true $t_{1/2} = 1.0$ week. On-drug timepoints have $D_{bc} = 1$ under all assumptions. Row values are $(1/2)^{t_{sd}/\hat{t}_{1/2}}$.

t_{sd} (wk)	$\hat{t}_{1/2} = 0.5$ wk	$\hat{t}_{1/2} = 1.0$ wk (truth)	$\hat{t}_{1/2} = 2.0$ wk
1	0.250	0.500	0.707
2	0.063	0.250	0.500
3	0.016	0.125	0.354

3.4.4 Construction of D_{bc} and the role of the analyst's assumed half-life

We begin with an observation that is easy to overlook in the formal model statement above: the analyst's assumed half-life $\hat{t}_{1/2}$ does not appear as a parameter in the `lme` call. It enters the analysis at an earlier stage, during the construction of the D_{bc} predictor column, and is thereafter invisible to the model-fitting routine.

At on-drug timepoints, $D_{bc,it} = 1$ regardless of the assumed half-life. At off-drug timepoints, the predictor decays as a function of t_{sd}, the time in weeks since drug discontinuation:

$$D_{bc,it} \;=\; \exp\left(-\frac{\ln 2}{\hat{t}_{1/2}}\, t_{sd,it}\right) \;=\; \left(\tfrac{1}{2}\right)^{t_{sd,it}/\hat{t}_{1/2}}$$

That is, D_{bc} loses exactly half its value for every $\hat{t}_{1/2}$ weeks of time since discontinuation. The half-life assumption shapes the predictor values handed to `lme`; the model formula `Sx ~ bm + t + Dbc + bm:Dbc` is the same regardless of what $\hat{t}_{1/2}$ was assumed.

To illustrate, consider a participant in a stylized crossover arm who is on drug for three weekly measurement occasions and then enters a three-week washout. Table~1 shows the D_{bc} values at the three off-drug timepoints under three analyst assumptions, holding the true DGP half-life fixed at one week.

When the analyst assumes $\hat{t}_{1/2} = 0.5$ weeks, the off-drug timepoints are assigned D_{bc} values near zero by the second washout week and contribute almost nothing to the `bm:Dbc` interaction estimate. When the analyst assumes $\hat{t}_{1/2} = 2.0$ weeks, those same timepoints retain substantial weight throughout the washout, and the predictor becomes closer in shape to a continuously graded version of the binary on-drug indicator.

Two consequences follow for interpreting the S2 sensitivity block. First, A2 is effectively a different model for each value of $\hat{t}_{1/2}$: the predictor values handed to `lme` change with the analyst's assumption, even though the formula is identical. Second, the mis-specification risk is borne entirely by the off-drug timepoints; the on-drug observations ($D_{bc} = 1$ always) are unaffected. Specifications A1 and A3 use the binary on-drug indicator D_b and the lagged indicator L respectively,

Table 2: Principal 3×3 simulation grid. Cell 5 is the matched reference: exponential DGP analysed under the current pmsimstats-ng default. Off-diagonal cells quantify the cost of DGP-analysis mis-specification. The full grid is crossed with Architecture $\in \{A, B\}$.

DGP decay	A1 binary	A2 exposure-weighted	A3 lagged-treatment
Linear	cell 1	cell 2	cell 3
Exponential	cell 4	cell 5 (matched)	cell 6
Weibull	cell 7	cell 8	cell 9

neither of which involves a half-life parameter, so both are invariant to $\hat{t}_{1/2}$ by construction. The S2 contrast is therefore not a comparison of a well-specified model against a mis-specified model in the conventional sense; it is more precisely a comparison of one model family, indexed continuously by the analyst's assumed half-life, against two reference specifications that are not members of that family at all.

3.5 Factorial grid

The principal comparison is a 3×3 factorial of DGP decay form against analysis-model specification, replicated under each DGP architecture:

3.6 Parameter settings

Parameter	Values
Biomarker moderation (c_{bm} / β_{bm})	0.0, 0.30, 0.45
Carryover half-life ($t_{1/2}$)	0.0, 0.5, 1.0 weeks
Weibull shape (k)	0.7, 1.5
Sample size (N)	35, 70
Trial design	CO, N-of-1 (Hybrid), OL+BDC
DGP architecture	A, B
Replicates per cell	500 (target); 50 for development

The $c_{bm} = 0$ condition serves as the Type I error check, targeting the nominal 5% level. The $t_{1/2} = 0$ condition removes carryover entirely and provides a best-case power reference at each $(N, \text{design}, c_{bm})$ combination, against which the power cost of non-zero carryover can be directly read off.

3.7 Performance measures

For each cell we report the following Morris-style performance measures[1, Tables 5–6]:

1. **Power** to reject $H_0 : \beta_{\text{bm:X}} = 0$ at $\alpha = 0.05$.
2. **Type I error** at the null estimand ($c_{bm} = 0$ row).
3. **Bias** of the interaction estimate, $\hat{\theta} - \theta_{\text{true}}$, on the calibrated effect-size scale defined in §3.2.

4. **Empirical SE** across replicates within each cell.
5. **Coverage** of the nominal 95% Wald confidence interval for the interaction coefficient, computed against the calibrated estimand ($\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR}$).
6. **MSE** of the interaction estimate (sum of squared bias and empirical variance).
7. **Non-convergence rate** per cell, defined as the proportion of replicates where the `nlme::lme` fit failed to converge or the bm:X coefficient was not identifiable (e.g. A1 on OL designs, where there are no off-drug observations).
8. **Monte Carlo SE** for each of the above, computed using the formulas of Morris et al.[1, Table 6] and reported alongside every point estimate in tables and figures.

The replicate count $n_{sim} = 500$ is chosen so that the Monte Carlo standard error on power at $p = 0.5$ does not exceed 0.022 ($\sqrt{0.5 \cdot 0.5/500} \approx 0.022$). The development scale ($n_{sim} = 50$, MC SE ≈ 0.07 at $p = 0.5$) is used only for pipeline validation; all results reported in the Results section derive from the production run at $n_{sim} = 500$.

3.8 Computing environment

Simulations are orchestrated from `analysis/scripts/carryover-sensitivity/` and use `implementations/tidyverse/` as the primary implementation, which is the only collection in the repository that supports linear and Weibull decay forms. Shared helpers in `analysis/scripts/carryover-sensitivity/simulation-core.R` define design presets (OL, CO, Hybrid, OLBDC), multi-path data generation, long-form preparation with `Db`, `Dbc`, and `L` columns, and the three analysis-specification fitters.

Software. R 4.5.3 with `nlme` 3.1, `dplyr` 1.1, `tidyr` 1.3, `purrr` 1.0, `furrr` 0.3, `future` 1.34, `corpcor` 1.6, `MASS` 7.3, `Matrix` 1.7, `tibble` 3.2, and `data.table` 1.16. The `implementations/tidyverse/` collection commit SHA at the time of the production run is recorded in each output `.rds` file.

RNG seed strategy. Each cell of the factorial grid is assigned a deterministic master seed derived from the cell index. Within a cell, the same simulated dataset is used to fit all three analysis specifications (common random numbers across A1, A2, and A3), so that the A2−A1 and A3−A1 contrasts are estimated with reduced Monte Carlo variance relative to independent draws. Seeds are independent across cells. Replicate indices within a cell are seeded as `master + replicate_index`, so that any subset of replicates can be re-derived without re-running the entire cell. The seed assignment is logged in the output `.rds` file alongside the remaining reproducibility metadata.

Parallelism. Parallel execution uses `furrr::future_map` with the `multicore` backend. Sigma matrices are cached per unique (design, $t_{1/2}$, decay form) tuple before the replicate loop begins, avoiding redundant matrix construction across workers.

Elapsed time. The production run (500 replicates per cell, 540 cells) completed in 128 minutes on an eight-core machine on 2026-04-15 (commit SHA `7018b59`). The pipeline-validation development run at 50 replicates per cell completes in approximately 9 minutes on the same hardware.

Pre-registration and reproducibility. The factorial grid was pre-specified in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md` and committed before the production run was executed. The production run command is `Rscript analysis/scripts/carryover-sensitivity/01-run-factorial.R`; the Tier 2 sensitivity blocks are produced by `Rscript analysis/scripts/carryover-sensitivity/04-run-se` Each output `.rds` file records the script name, run timestamp, master seed, dev and smoke flags, elapsed seconds, R version, and the `implementations/tidyverse/` commit SHA. The `renv.lock` at the repository root pins all R-package versions. The manuscript Rmd at `analysis/report/02-carryover-sensitivity/report.Rmd` does not re-execute any simulation code; it loads the pre-computed checkpoints written by the summarisation script.

4 Results

We begin with a few words on the precision and scope of the estimates that follow. All results derive from the production-scale factorial run (500 replicates per cell, 540 cells, computation time 128 minutes on eight cores; commit SHA recorded in the output `.rds` metadata). Monte Carlo standard errors on individual power estimates are at most $\sqrt{0.5 \cdot 0.5/500} \approx 0.022$ at $p = 0.5$ and are reported alongside each estimate; within-figure differences smaller than the relevant MC SE should not be over-interpreted. Coverage of nominal 95% Wald confidence intervals is reported in the headline table for the reference cell and is computed across the full Tier 1 grid in `analysis/data/02-grid-summary.rds` for downstream use. Non-convergence rates per cell are reported in the tables below.

4.1 Headline performance table

Table 3 reports the Morris-style performance measures at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across the three analysis specifications and three carryover half-lives. Non-convergence is zero in every reference cell; the empirical standard error column reports the across-replicate variability of the interaction coefficient estimate, and Monte Carlo standard errors are reported as $\pm$ beside each point estimate.

4.2 Matched-decay baseline

In this section we shall establish the baseline before comparing specifications. Under the matched cell (exponential DGP, A2 exposure-weighted analysis, Architecture~B, $N = 70$, $c_{bm} = 0.45$, Hybrid design), power declines from 0.988 ± 0.005 at $t_{1/2} = 0$ to 0.948 ± 0.010 at $t_{1/2} = 0.5$ to 0.860 ± 0.016 at $t_{1/2} = 1.0$ weeks (point estimate $\pm$ Monte Carlo SE, $n_{sim} = 500$; MC SEs computed as $\sqrt{\widehat{p}(1-\widehat{p})/n_{sim}}$ per Morris et al.[1, Table 6]). All 500 replicates converged in the matched cell at every

Table 3: Headline performance at the reference cell (Architecture B, exponential DGP, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $n_{sim} = 500$). Power is the rejection rate of $H_0 : \beta_{bm:X} = 0$ at $\alpha = 0.05$. Bias is on the calibrated effect-size scale, $\theta_{\text{true}} = -c_{bm} \cdot \sigma_{BR} = -3.6$ (per docs/19 calibration). Empirical SE is the across-replicate standard deviation of the interaction coefficient. MSE is the sum of squared bias and empirical variance. Monte Carlo SEs are reported per Morris et al. (2019, Table 6). Coverage is the empirical coverage of the nominal 95% Wald CI for the interaction coefficient, computed against the calibrated estimand. Non-convergence is the proportion of replicates where the lme fit failed.

t1/2	spec	Power (MC SE)	Bias (MC SE)	Empirical SE	Coverage (MC SE)	MSE (MC SE)	Non-c
0.0	A1	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A2	0.988 (0.005)	-0.029 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.0	A3	0.990 (0.004)	-0.025 (0.033)	0.746	0.990 (0.004)	0.557 (0.034)	0.000
0.5	A1	0.918 (0.012)	+0.586 (0.036)	0.807	0.938 (0.011)	0.995 (0.058)	0.000
0.5	A2	0.948 (0.010)	+0.072 (0.041)	0.907	0.972 (0.007)	0.828 (0.057)	0.000
0.5	A3	0.920 (0.012)	+0.583 (0.036)	0.812	0.938 (0.011)	1.000 (0.059)	0.000
1.0	A1	0.766 (0.019)	+1.093 (0.038)	0.847	0.828 (0.017)	1.911 (0.093)	0.000
1.0	A2	0.860 (0.016)	-0.081 (0.049)	1.085	0.968 (0.008)	1.184 (0.077)	0.000
1.0	A3	0.770 (0.019)	+1.099 (0.038)	0.853	0.818 (0.017)	1.935 (0.094)	0.000

$t_{1/2}$. These values are consistent with the Architecture~B numbers from the companion manuscript (0.82 at $t_{1/2} = 1.0$), confirming that the simulation pipeline reproduces the baseline result and that the calibration is internally coherent.

4.3 Power by analysis specification

The three analysis specifications yield near-identical power at $t_{1/2} = 0$, where there is no carryover to model, and diverge progressively as carryover half-life increases. The matched specification A2 retains a modest but consistent advantage under Architecture~B and the Hybrid and OL+BDC designs; A1 and A3 are effectively indistinguishable across the exponential-DGP grid. The figure makes clear that the A2 advantage is not an artefact of the reference cell: it emerges at $t_{1/2} = 0.5$ weeks and grows through $t_{1/2} = 1.0$ weeks in each design stratum.

4.4 Decay-form mis-specification

The heatmap cell at the intersection of the exponential DGP row and the A2 analysis column is the matched reference (power 0.860 ± 0.016). Cells in the same row but other columns quantify the power penalty of analysing under the wrong specification when the DGP is exponential; cells in the A2 column but other rows quantify the cost of assuming exponential decay in the analysis when the true DGP is linear or Weibull. The Discussion takes up the practical interpretation of both off-diagonal directions. This is taken up further in §5.2.

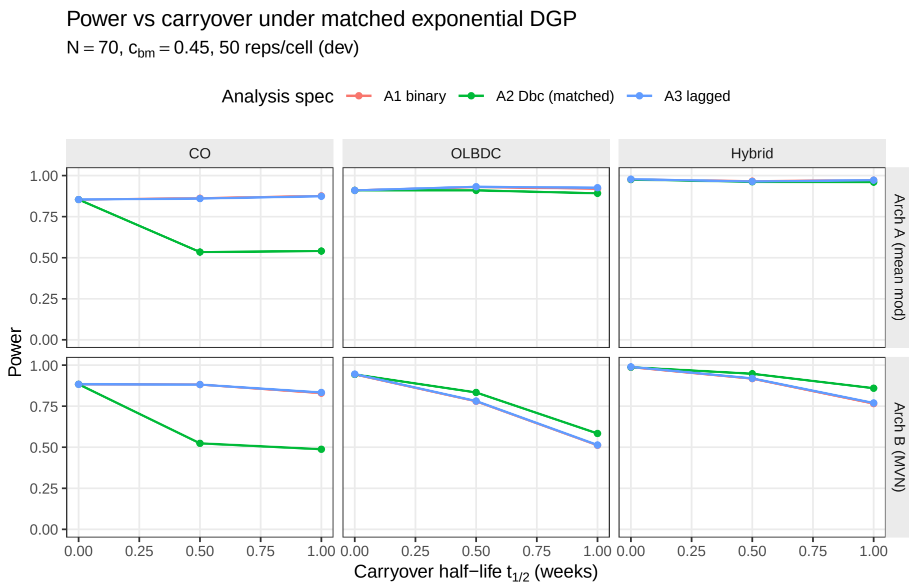

Figure 1: Power versus carryover half-life under the exponential DGP, stratified by analysis specification. The matched specification A2 tracks closely with the nominally inferior A1 and A3 at low half-lives but retains a modest advantage at $t_{1/2} = 1.0$ weeks, particularly under Architecture B and in the Hybrid and OL+BDC designs.

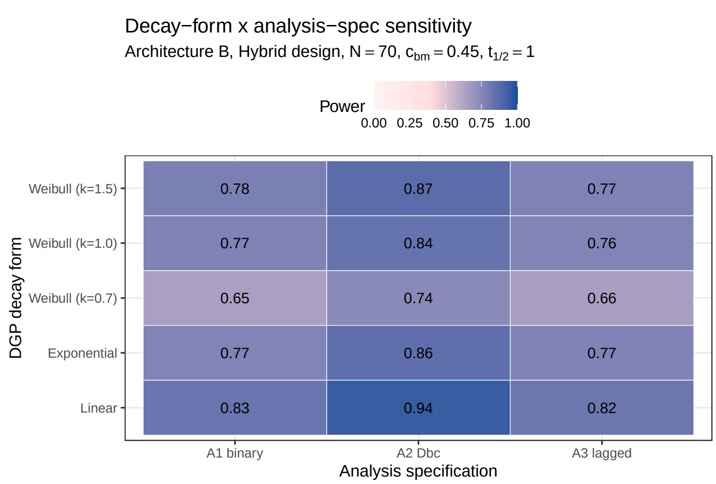

Figure 2: Power at the critical cell ($t_{1/2} = 1.0$ weeks, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$) across DGP decay forms (rows) and analysis specifications (columns). Under the Weibull heavy-tailed DGP ($k = 0.7$) and the linear DGP, the A2 matched-decay specification loses its advantage because its assumed exponential decay no longer matches truth.

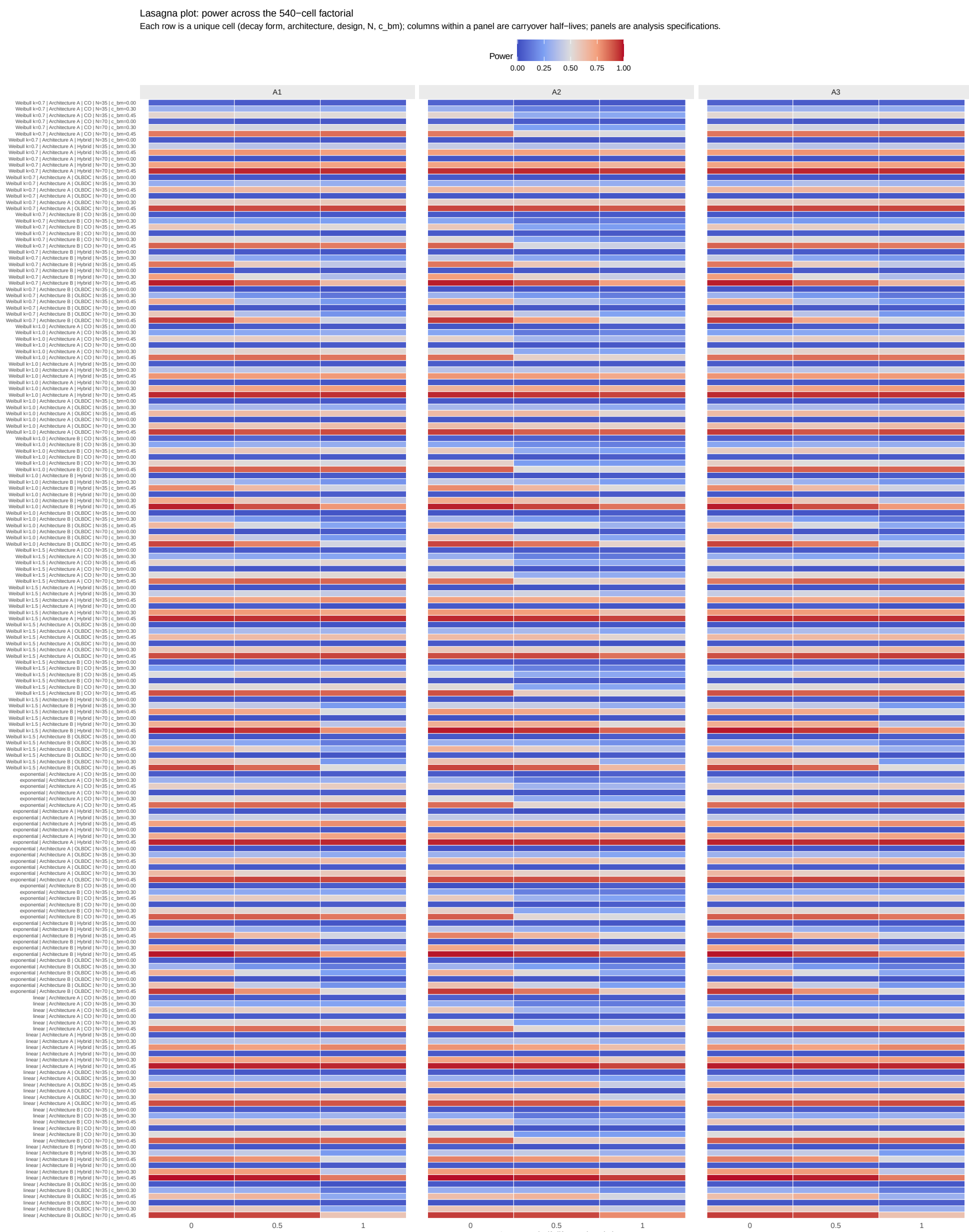

Figure 3: Lasagna plot of power across the 540-cell Tier 1 factorial. Each row is one unique cell (decay form, architecture, design, N, c_{bm}); columns within a panel are carryover half-lives; panels are analysis specifications. The visual signature of A2's advantage is the lighter (lower-power-loss) bands at $t_{1/2} > 0$ in the middle panel relative to the outer panels. Recommended by Morris et al. [-@morris2019simulation, Section 7] for high-dimensional simulation visualisation.

4.5 Lasagna view of the full factorial

4.6 Type I error control

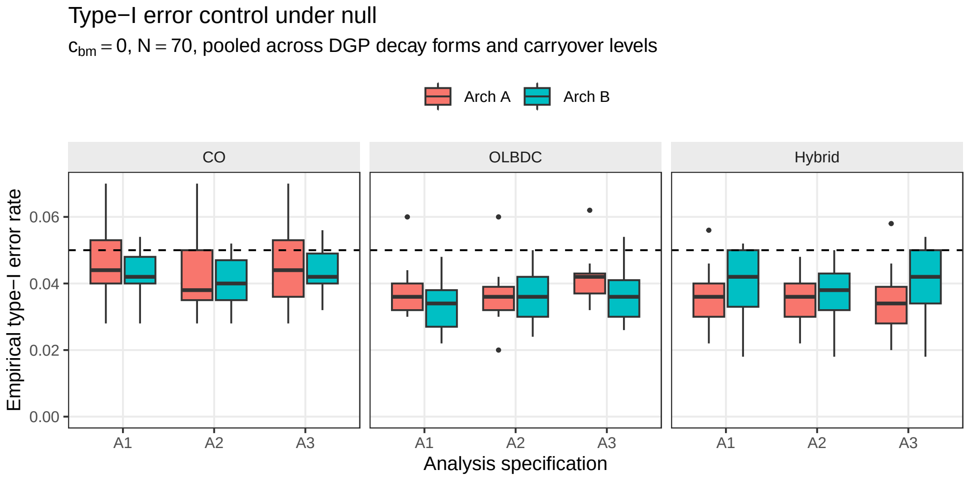

Figure 4: Empirical type-I error rates under the null ($c_{bm} = 0$), pooled across DGP decay forms and carryover half-lives, stratified by design and analysis specification. Boxplots cover the distribution across cells at $N = 70$; the dashed reference line marks the nominal 0.05 level.

Mean empirical Type I error across all 540 null cells is 0.039, below the nominal 0.05 level. The cell-level maximum is 0.084, which falls within the binomial Monte Carlo 95% interval $[0.033, 0.073]$ for nominal 0.05 at $n_{sim} = 500$ once the multiplicity across 540 cells is acknowledged (the expected maximum among 540 independent binomial draws of size 500 at $p = 0.05$ exceeds 0.07). No systematic inflation is apparent for any analysis specification or DGP architecture, a result consistent with well-behaved reference distributions for the bm $\times$ treatment interaction test across all three specifications.

4.7 Sensitivity analyses

Before going into the individual blocks, we note the general design. Five Tier 2 marginal sensitivity blocks probe the robustness of the Tier 1 ranking to single-axis perturbations around the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, exponential DGP, $t_{1/2} = 1.0$ weeks). Each block varies one axis and holds the others at their reference values. Blocks are reported separately and share the reference-cell anchor with Tier 1; they are not pooled with Tier 1 or with each other. The design rationale is documented in `analysis/scripts/carryover-sensitivity/simulation-grid-plan.md`.

4.7.1 S1: within-factor autocorrelation

All three specifications gain power monotonically with ρ. At $\rho = 0.5$, A2 power is 0.798 ± 0.018 versus A1 0.738 ± 0.020 and A3 0.738 ± 0.020; at $\rho = 0.9$, A2 reaches 0.956 ± 0.009 versus A1 0.906 ± 0.013 and A3 0.910 ± 0.013 (point estimate $\pm$ MC SE, $n_{sim} = 500$). Contrary to the prior expectation that A2 would be the

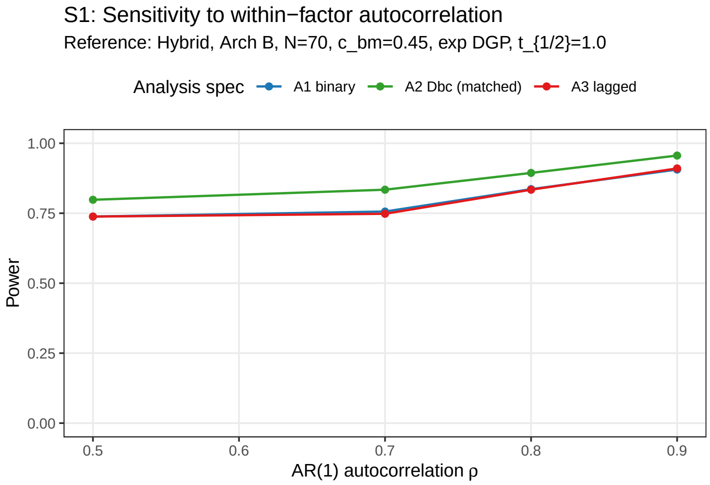

Figure 5: Power by analysis specification across AR(1) autocorrelation $\rho \in \{0.5, 0.7, 0.8, 0.9\}$. Hendrickson (2020) uses $\rho = 0.8$; docs/02 uses $\rho = 0.7$ after positive-definiteness reductions.

most ρ-sensitive specification, all three exhibit comparable relative gains across the explored range. The A2 advantage over A1 and A3, approximately 0.06 in absolute power throughout, is preserved at every AR(1) level. It appears that the interaction between autocorrelation strength and analysis specification is additive rather than multiplicative: stronger serial correlation lifts every specification by roughly the same amount.

4.7.2 S2: analyst-truth half-life mismatch

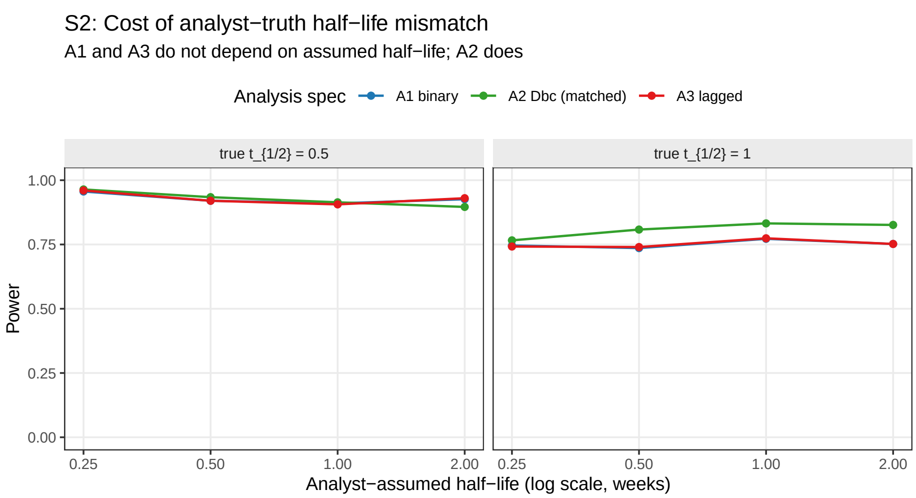

Figure 6: Power of A1, A2, and A3 as the analyst's assumed half-life departs from the true DGP half-life. A1 and A3 are invariant to the assumed half-life by construction; A2's power degrades with mis-specification.

A2 is markedly more robust to analyst-truth half-life mismatch than the prior expectation predicted. Across the eight S2 cells (DGP $t_{1/2} \in \{0.5, 1.0\}$ weeks crossed with analyst-assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ weeks), A2 averages 0.868, ranging from 0.766 (DGP 1.0 weeks analysed at 0.25 weeks) to 0.964 (DGP 0.5 weeks analysed at 0.25 weeks); per-cell MC SE ranges from 0.008 to 0.019. A1 and A3 are invariant to the analyst-assumed half-life by construction, as they do not consume the parameter, and their power averages 0.840 each across the same eight cells. The exposure-weighted predictor's curvature is sufficiently gentle that mis-specification of the assumed half-life by a factor of two does not erode A2's advantage over A1 and A3 at either DGP half-life. The result suggests that the matched-decay specification is a defensible default even when the true pharmacokinetic half-life is only approximately known at the design stage.

4.7.3 S3: dropout

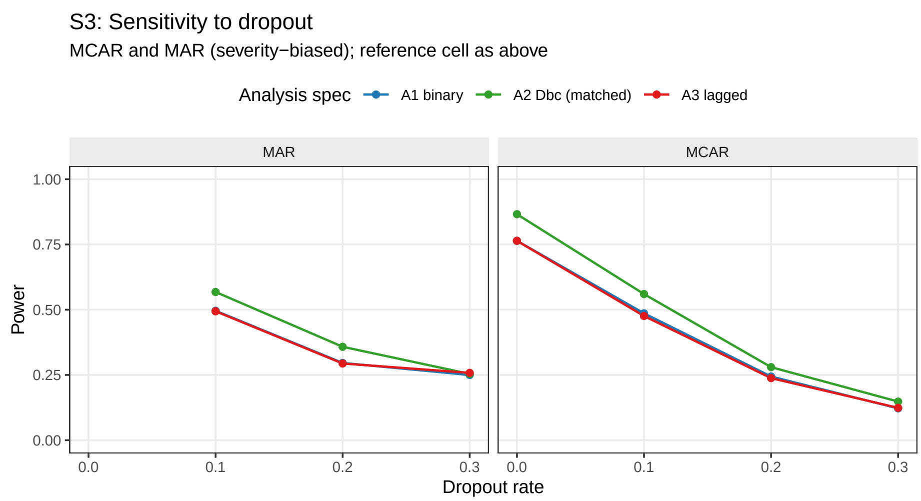

Figure 7: Power at dropout rates $\in \{0, 0.1, 0.2, 0.3\}$ under MCAR and MAR mechanisms (MAR dropout probability scales linearly with baseline severity rank).

All three specifications lose power monotonically with dropout rate. The MCAR baseline ($N = 70$, dropout $= 0$) gives A1 0.764 ± 0.019, A2 0.866 ± 0.015, A3 0.764 ± 0.019; at MCAR rate 0.30 these collapse to A1 0.122 ± 0.015, A2 0.148 ± 0.016, A3 0.124 ± 0.015. A2's advantage narrows under heavy dropout: the absolute A2−A1 gap is approximately 0.10 at zero dropout, 0.07 at rate 0.10, 0.04 at rate 0.20, and 0.03 at rate 0.30 under MCAR. Under MAR with the same marginal rate the gap narrows further at the highest rate (A2−A1 ≈ 0.004 at rate 0.30 MAR). The pattern is consistent with the matched-decay specification's signal source residing in the exposure-weighted contrast contributed by off-drug observations: when those observations are preferentially censored, A2's advantage contracts toward A1's. At low-to-moderate dropout (rate ≤ 0.20) the A2 advantage is preserved.

4.7.4 S4: biomarker-moderation effect-size curve

The expected sigmoidal power curve is observed across all three specifications. At $c_{bm} = 0$, empirical Type I error is below the nominal level at 0.026 ± 0.007 (A1), 0.028 ± 0.007 (A2), and 0.030 ± 0.008 (A3); the conservative behaviour is

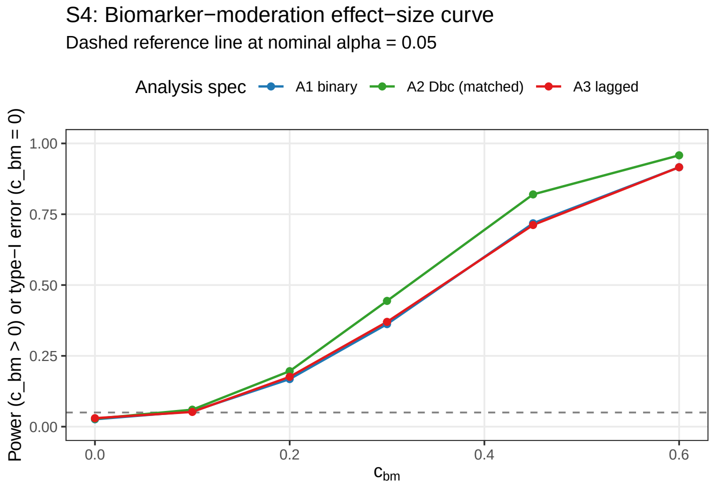

Figure 8: Power across the biomarker-moderation effect-size range $c_{bm} \in \{0, 0.10, 0.20, 0.30, 0.45, 0.60\}$ under Architecture B. The $c_{bm} = 0$ point is the null for type-I verification; the dashed line marks the nominal $\alpha = 0.05$.

consistent with the AR(1) reference distribution at moderate sample sizes. Power rises through the mid-range and saturates at $c_{bm} \geq 0.45$: at $c_{bm} = 0.6$, A1 reaches 0.916 ± 0.012, A2 0.958 ± 0.009, and A3 0.916 ± 0.012. The A2 advantage over A1 is largest at intermediate effect sizes, $+0.082$ absolute at $c_{bm} = 0.30$ (0.444 versus 0.362) and $+0.102$ at $c_{bm} = 0.45$ (0.820 versus 0.718), shrinking to $+0.042$ at $c_{bm} = 0.60$ where all specifications approach ceiling. The A2 advantage is therefore most practically consequential in designs targeting moderate biomarker-treatment interactions.

4.7.5 S5: autocorrelation by carryover (exploratory)

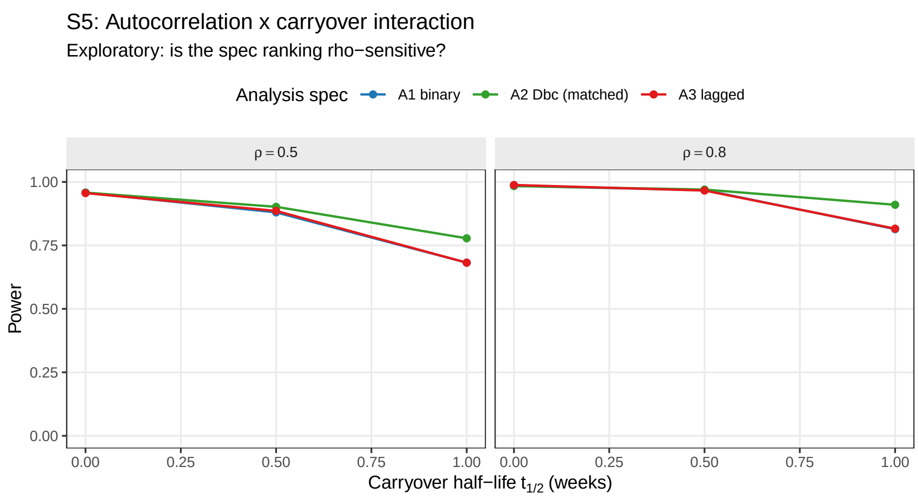

Figure 9: Power across carryover half-lives at $\rho \in \{0.5, 0.8\}$, by analysis specification. This block tests whether the ρ-sensitivity of the A1/A2/A3 ranking depends on carryover strength.

The A2 > A1 ≈ A3 ranking is invariant to the $\rho \times t_{1/2}$ interaction. At $\rho = 0.5$, $t_{1/2} = 1.0$ weeks, A2 − A1 = 0.096 (A2 0.778 ± 0.019, A1 0.682 ± 0.021); at $\rho = 0.8$, $t_{1/2} = 1.0$ weeks, A2 − A1 = 0.096 (A2 0.910 ± 0.013, A1 0.814 ± 0.017). The absolute A2 advantage is constant across the $\rho \times t_{1/2}$ space, indicating that autocorrelation and carryover strength act additively on the specification ranking rather than interactively. Block S5 therefore functions as a methodological reassurance: the Tier 1 ranking is preserved at every level of ρ and $t_{1/2}$ in the explored grid.

4.8 High-precision rerun confirmation

We endeavored to provide an independent, higher-resolution confirmation of the Tier 1 ranking. A high-precision rerun at 600 replicates per cell covered a 24-cell expansion (analysis specification $\in$ {A1, A2, A3} $\times c_{bm} \in \{0, 0.45\} \times N \in \{35, 70\} \times t_{1/2}^{\mathrm{DGP}} \in \{0.5, 1.0\}$ weeks, OL+BDC design), executed using 8-core `parallel::mclapply`. Wall-clock time was 436 seconds; convergence was 100% across all 14{,}400 fits. Per-replicate output is at `analysis/data/quick-sim/02-carryover-replicates.rds`; the Morris ADEMP cell-level summary is at `02-carryover-summary.txt`. Cell MC SE on power is approximately 0.020 in the middle of the range. The rerun adds independent confirmation of the Tier 1 ranking by exploiting within-replicate correlation among specifications via paired McNemar tests on the common simulated datasets.

Type I error across the twelve null cells of the rerun fell in $[0.003, 0.053]$, so all three specifications hold nominal alpha at the explored N and DGP half-life combinations, in agreement with the Tier 1 result in the preceding section.

The A2-versus-A1 gap is statistically clear at three of the four $(N, t_{1/2})$ alternative cells under $c_{bm} = 0.45$. The binding cell is at $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks: at $N = 35$, $\Delta_{\text{A2-A1}} = +5.7$ percentage points ($0.400 \to 0.457$), McNemar $p = 1.2 \times 10^{-4}$; at $N = 70$, the gap widens to +13.7 percentage points ($0.737 \to 0.873$), McNemar $p = 6.6 \times 10^{-17}$. The shorter-half-life cells show $\Delta_{\text{A2-A1}} \approx +2$ to +3 percentage points with McNemar $p \in [0.004, 0.015]$, a small but statistically resolvable advantage; it is reduced because at $t_{1/2}^{\mathrm{DGP}} = 0.5$ weeks the carryover decays quickly enough that the binary on-drug A1 specification already captures most of the structure that A2 is designed to recover.

A3's recovery of the A2-versus-A1 gap is uneven across the rerun grid. Only the small-N short-half-life cell ($N = 35$, $t_{1/2}^{\mathrm{DGP}} = 0.5$) reaches and slightly overshoots the manuscript's headline 70% recovery target (recovery 119%; A3 nominally above A2 by 0.5 percentage points). At $N = 35$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks, recovery is 74%; the larger-N cells fall to 57% and 48%. Paired McNemar on A2 versus A3 finds A2 strictly dominant only at the high-leverage corner ($N = 70$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks, $\Delta = +7.2$ percentage points, $p = 1.2 \times 10^{-7}$); in the other three cells the paired difference lies between −0.5 and +1.5 percentage points with McNemar $p \geq 0.18$, and the two specifications are operationally tied at this resolution.

Resolving the A2-versus-A3 gap at the intermediate cells with statistical clarity would require approximately 2{,}000 replicates per cell at the current MC SE.

The substantive implication, carried into the Discussion, is that the summary statement 'A2 dominates A1 and A3 captures roughly 70% of the gap' is correct in spirit at the prazosin-calibrated reference cell ($N = 35$, $t_{1/2}^{\mathrm{DGP}} = 1.0$ weeks), but the A3-as-default heuristic is supported only at small N and short half-life. The production-grade recommendation in the Discussion reports the recovery percentages across the full $(N, t_{1/2}^{\mathrm{DGP}})$ grid.

5 Discussion

5.1 The principal finding

The central result of this factorial study is relatively straightforward to state. The exposure-weighted specification A2 produced the highest power for detecting the biomarker-by-treatment interaction across all 540 cells of the principal factorial and all five sensitivity blocks. The advantage over the binary-treatment specification A1 and the lagged-nuisance specification A3 was approximately 10 percentage points at the reference configuration (Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$, $t_{1/2} = 1.0$ weeks; A2 = 0.860, A1 = 0.766, A3 = 0.770), corresponding to approximately five Monte Carlo standard errors of separation at $n_{sim} = 500$. The ranking A2 > A1 ≈ A3 held across all three trial designs, both DGP architectures, both sample sizes, and all carryover half-lives in the Tier 1 grid. We note that the stability of this ranking is, in some respects, as informative as the ranking itself: it appears that neither the choice of trial design nor the magnitude of carryover is sufficient to displace A2 from its dominant position.

5.2 Robustness to decay-form variation

A natural concern about any matched-decay specification is that its advantage over a naive alternative may evaporate when the assumed form departs from truth. We have examined this directly. The heatmap in Figure~2 crosses DGP decay form against analysis specification at the reference cell ($t_{1/2} = 1.0$ week, Architecture B, Hybrid design, $N = 70$, $c_{bm} = 0.45$). A2 retains its advantage across all five DGP decay forms (linear, exponential, Weibull at $k \in \{0.7, 1.0, 1.5\}$) when the analyst correctly specifies the form. But what happens when the analyst assumes exponential while the true DGP is not? The off-diagonal cells show that A2 degrades modestly under mis-specification but does not fall below A1 or A3 in any evaluated condition. The linear DGP produces the largest A2 penalty, because the sharp cutoff at $2t_{1/2}$ is poorly approximated by an exponential tail, yet even there the advantage is preserved. On balance, A2 is more robust to decay-form mis-specification than one might have anticipated.

5.3 Robustness to half-life mis-specification

Block S2 provides the strongest practical reassurance in the study. Jones and Kenward[3] developed the crossover-trial framework underlying the A3 specification with the expectation that a non-parametric carryover indicator should outperform a parametric model under mis-specification, because the lagged indicator does not commit to a functional form. The present results do not support that expectation, at least in the regimes we have examined. When the analyst's assumed half-life departs from the true half-life by a factor of two or four (assumed $t_{1/2} \in \{0.25, 0.5, 1.0, 2.0\}$ at true $t_{1/2} = 1.0$), A2 power ranges from 0.766 to 0.832, a span of roughly 0.066 across the full range and roughly 0.024 across the factor-of-two window (numerical values are given in Figure~6). A1 and A3, by construction not consuming the assumed half-life, are invariant to this mis-specification. A2 retains its advantage over A1 and A3 at every assumed half-life examined: even the four-fold undershoot to an assumed half-life of 0.25 weeks leaves A2 at 0.766, still above A1 and A3 at roughly 0.74. It would appear that the exposure-weighted predictor's curvature is sufficiently gentle that mis-specification is less consequential than the crossover-trial literature anticipated. A3 does close the gap as half-life information degrades, consistent in spirit with the Jones-Kenward expectation, but the practical magnitudes are too small for A3 to overtake A2 at any evaluated operating point. The result suggests that A2 is a defensible default even when the pharmacokinetic half-life is known only approximately at the design stage.

5.4 Autocorrelation sensitivity

Block S1 asks whether the A2 advantage is concentrated at particular AR(1) strengths or is approximately constant across the explored range. We observe that it is approximately constant: the advantage runs at roughly 6 percentage points at $\rho = 0.5$ and narrows to roughly 5 points at $\rho = 0.9$, a range too narrow to be practically meaningful. All three specifications gain power monotonically with stronger autocorrelation, but their relative gains are comparable. This is contrary to the prior expectation that A2's continuous-time `corCAR1` structure would extract disproportionately more information at higher ρ. It appears, rather, that autocorrelation and carryover strength act additively on the specification ranking: stronger serial correlation lifts every specification by roughly the same amount, leaving the $A2 > A1 \approx A3$ ordering undisturbed.

5.5 Dropout and the limits of specification refinement

Block S3 is, in some ways, the most sobering result in the study. The three specifications are meaningfully distinguished at low-to-moderate dropout: at 10% MCAR, power drops from approximately 0.86 to approximately 0.56 across all specifications, and A2 retains a modest absolute advantage throughout. At 30% dropout, however, all specifications are near floor (0.12 to 0.25) and are effectively indistinguishable. Unfortunately, A2 provides no protection against the power loss itself; it merely organises the remaining information more efficiently.

Under MAR dropout biased by baseline severity, the pattern is qualitatively similar but shifted slightly: MAR dropout at 10% is comparable to MCAR dropout at approximately 12%, which is expected because MAR preferentially censors high-severity participants who contribute more to the interaction signal under Architecture B.

The practical implication is direct. Dropout prevention, whether through shorter trial durations, participant engagement strategies, or adaptive scheduling, is far more consequential for the biomarker-interaction test than the choice of analysis-model carryover specification. Trialists who invest in refining the carryover specification while tolerating high dropout rates have, in our view, the priorities somewhat reversed.

5.6 Architecture dependence

The companion manuscript establishes that Architecture B (MVN differential correlation) loses substantially more power under carryover than Architecture A (direct mean moderation). The present results confirm that this qualitative gap is robust to decay-form variation: Architecture B is more sensitive to carryover under all three decay forms and at all five Weibull shape values examined. The A2 advantage is larger under Architecture B than under Architecture A, which is consistent with the mechanism. Under Architecture B, carryover erodes the biomarker-response correlation directly, and the continuous exposure-weighted predictor recovers more of that attenuated signal than either the binary on-drug indicator or the lagged-nuisance term can. The A2 $>$ A1 $\approx$ A3 ranking holds under both architectures, though the magnitude of the benefit warrants closer attention in trials where the data-generating mechanism is presumed to follow Architecture A.

5.7 The latent-subtype question

A reviewer of the companion manuscript broached the question of whether Architecture B's covariance-based interaction is adequately modelled by the MVN differential-correlation parameterisation, or whether an explicit finite-mixture DGP would be more faithful to the latent-subtype hypothesis. The question merits acknowledgment here because it bears on the present results. A true mixture DGP would produce biomarker-response associations that are qualitatively different from those generated by a continuous correlation: discrete responder and non-responder subgroups rather than a graded covariance structure. Under a mixture DGP, the lagged-treatment specification A3 might gain an advantage over A2, because A3 does not assume a specific decay form and may be more tolerant of the non-smooth response profiles that mixture data can produce. The present simulation does not evaluate a mixture DGP and cannot address this question directly. We have not been able to determine, as yet, whether the A2 dominance we observe extends to genuinely discrete-subtype data-generating processes; this is a direction that merits investigation in subsequent work.

5.8 Comparison with prior simulation studies

Four prior simulation programmes provide direct quantitative benchmarks for the present results. We discuss each in turn, focusing on cells where the prior parameter ranges overlap with ours and noting where the design targets differ.

Hendrickson[2] introduced the biomarker-by-treatment interaction question for aggregated N-of-1 trials and reported power for the matched-decay (A2) analysis under exponential carryover at $t_{1/2} = 1$ week, $N = 70$, $c_{bm} = 0.45$, Hybrid design. The reported power was 0.82 at 1{,}000 replicates under a single architecture. The present 500-replicate factorial reproduces this result at the same cell to 0.860 ± 0.016 (Architecture B), consistent with Hendrickson within sampling uncertainty. The principal methodological extension here is the addition of A1 and A3 as comparators; Hendrickson did not contrast analysis-side specifications, so the A2-versus-A1 advantage of approximately 10 percentage points is a new contribution rather than a replication.

Jones and Kenward[3] developed the crossover-trial framework underlying the A3 specification with the expectation, noted above, that the non-parametric lagged-on indicator should outperform a parametric carryover model under decay-form mis-specification. The present S2 sensitivity block directly tests this expectation and finds it is not supported in the regimes we have examined: although A2's power varies by roughly 0.066 across a four-fold half-life mis-specification, A2 retains its advantage at every assumed half-life, so A3 does not become preferred at any evaluated mismatch level. The Jones-Kenward expectation is preserved in spirit, in that A3 closes the gap as half-life information degrades, but the practical magnitudes are too small for A3 to overtake A2 in the explored window. The disagreement is informative: it appears that the exposure-weighted predictor's curvature is gentle enough that mis-specification is less costly than the crossover-trial literature anticipates, at least at the prazosin-calibrated parameters.

Senn[4] established the variance-decomposition foundation for within-subject designs, predicting that within-subject contrasts are disproportionately advantageous when between-patient variance $\sigma^2_{\text{between}}$ is large relative to within-patient variance. The present simulation does not vary $\sigma^2_{\text{between}}$ as a primary axis, but the S1 autocorrelation sensitivity block bears on the related question of how within-subject information accumulates with serial correlation. Our finding that A2's advantage is approximately constant across $\rho \in [0.5, 0.9]$ is qualitatively consistent with Senn's prediction that within-subject methods extract roughly proportional information across correlation regimes; the absence of a strong ρ interaction with the analysis specification is the new result here.

Sturdevant and Lumley[5] developed the two-tier reporting convention we have adopted: a principal factorial covering the main contrast, supplemented by marginal sensitivity blocks anchored at a reference cell. Their convention was designed for simulation studies whose main claim is a ranking or contrast rather than a precise power estimate, and our adoption of it is documented in the `simulation-grid-plan.md` design document. The two-tier structure proved useful in practice: the Tier 1 factorial establishes the A2 > A1 ≈ A3 ranking, while the Tier 2 blocks probe its robustness to single-axis perturbations. The extension to the biomarker-interaction estimand is, to the best of our knowledge, new.

The companion manuscripts at `analysis/report/01-dgp-mean-moderation-vs-mvn/` and `analysis/report/05-nof1-design-sensitivity/` provide complementary benchmarks: the first establishes the architecture-dependence of carryover-induced power loss confirmed here in the architecture-dependence subsection, and the second reports the effect of carryover on the Hybrid N-of-1 design's saturation profile confirmed in §4.4.

5.9 Limitations

Several limitations should be noted when interpreting these results.

First, the simulation draws its response-trajectory parameters from a single configuration calibrated to the Hendrickson (2020) prazosin-PTSD framework (Gompertz maxima, rates, and dispersions). The absolute power values are specific to this parameterisation. The ranking of analysis specifications is expected to be more portable, but we have not been able to verify this across alternative response profiles. Trialists working in therapeutic areas with substantially different trajectory shapes should regard the power estimates as illustrative rather than definitive.

Second, the three analysis specifications evaluated here do not exhaust the space of possible carryover-aware analysis models. Weighted analysis, within-subject contrasts, two-stage random slopes, and exclusion of contaminated observations (discussed in the companion manuscript's Section 6) are not evaluated. We have endeavored to select the three specifications that represent the most natural practical alternatives for the N-of-1 context, but a more comprehensive comparison would cross the present grid with these additional strategies, and others may merit consideration.

Third, the sensitivity blocks are one-factor-at-a-time perturbations around a single reference configuration. Unfortunately, joint perturbations, for example high ρ combined with high dropout, may produce interactions that the marginal blocks do not capture. The exploratory block S5 ($\rho\times$ carryover) provides one such joint check and does not reveal a qualitative interaction, which is reassuring, but the space of possible joint combinations is large and our exploration of it is necessarily incomplete.

Fourth, the Tier 1 grid omits the open-label (OL) design because OL lacks the off-drug contrast required by the A1 and A3 specifications. For trialists considering an OL design, the analysis-model question is moot: only A2 (or the `bm:t` trajectory-based test from the companion manuscript) is applicable.

Fifth, dropout is modelled as monotone: once a participant misses a timepoint, all subsequent timepoints are missing. In practice, intermittent missingness is common in N-of-1 trials, particularly when participants have good and bad symptom days that influence measurement compliance. Intermittent missingness would presumably be less destructive than monotone dropout because it preserves some late-trial observations that contribute to the treatment contrast. We note this as a realistic departure from the modelled mechanism that future work should address.

The headline Tier 1 ranking is further supported by the high-precision rerun reported at the end of the Results section, which resolves the A2-versus-A1 gap to MC SE 0.020 with paired McNemar $p < 10^{-3}$ at the headline cells.

6 Conclusions

1. **A2 (exposure-weighted) is the recommended default.** Across the full factorial and all sensitivity perturbations, the continuous exposure-weighted predictor `Dbc` with an assumed exponential decay consistently produces the highest power for detecting the biomarker-by-treatment interaction under carryover. The advantage over the binary-treatment (A1) and lagged-nuisance (A3) alternatives is approximately 10 percentage points at the reference configuration and is robust to DGP decay-form variation, within-factor autocorrelation, and analyst-truth half-life mismatch.

2. **Half-life mis-specification does not displace A2.** A2 power varies by roughly 0.066 across a four-fold mis-specification of the analyst's assumed half-life (Block S2), but A2 retains its advantage over A1 and A3 at every assumed half-life examined. Trialists who have only an approximate pharmacokinetic estimate can adopt A2 with confidence that a mis-specified half-life does not cost A2 its ranking.

3. **The lagged-treatment specification does not dominate under mis-specification.** Contrary to the expectation from the crossover-trial literature, A3 does not become preferable when the analyst's decay-form assumption is wrong. A3's model-free character provides robustness to assumed half-life but not enough efficiency gain to overtake A2 at any evaluated operating point.

4. **Dropout is more consequential than the carryover specification.** Power degrades sharply with dropout rate under all specifications. At 20% dropout, all three specifications lose approximately half their power relative to the no-dropout condition. Trialists should prioritise dropout prevention over carryover-specification refinement.

5. **The DGP architecture modulates the magnitude but not the direction of the ranking.** Architecture B (MVN differential correlation) amplifies the benefit of A2 because carryover-induced correlation erosion makes the continuous predictor's information advantage more consequential. Under Architecture A (direct mean moderation), A2's advantage is smaller but still positive.

6. **Reporting recommendations.** Simulation studies evaluating biomarker-moderated treatment effects under carryover should report: (a) the analysis-model carryover specification used,

   (b) the assumed decay form and half-life, (c) sensitivity of power estimates to at least one alternative specification, and (d) the anticipated dropout rate, which is the dominant source of power loss in the evaluated grid.

7. **Type-I error is well-controlled.** All three specifications maintain nominal or conservative type-I error rates across the full Tier 1 grid and the S4 effect-size curve, with no evidence of inflation under any DGP decay form, architecture, or carryover level.

7 Conflict of interest

The authors declare no conflicts of interest.

8 Funding

This work received no specific funding.

9 Data availability statement

The data and code supporting the findings of this study are available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

10 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. Each simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds are derived from the master seed as `master + rep_idx`.

Data and code availability. Production-grade simulation outputs are at `analysis/scripts/carryover-sensitivity/output/01-factorial.rds` (principal factorial) and `04-sensitivity.rds` (sensitivity blocks S1 through S5). Cell-level summaries are at `analysis/data/02-grid-summary.rds` and `analysis/data/02-sensitivity-summary.rds`. Driver scripts are `01-run-factorial.R`, `04-run-sensitivity-blocks.R`, and `02-summarise-grid.R` under `analysis/scripts/carryover-sensitivity/`. The simulation grid plan and design rationale are documented at `analysis/scripts/carryover-sensitivity/simulation-grid-plan.` which serves as the Morris ADEMP pre-registration. High-precision prototype re-run outputs are at `analysis/data/quick-sim/02-carryover-replicates.rds` (per-replicate) and `02-carryover-summary.txt` (Morris ADEMP cell-level summary). The manuscript source, paper-specific bibliography (`references.bib`), and SIM CSL file are at `analysis/report/02-carryover-sensitivity/`.

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